

Torsion in Nil K -groups of schemes

Abstract

Let X be a Noetherian quasi-excellent scheme and let p be a prime such that $X[1/p]$ is regular. For every integer N , one power of p annihilates its polynomial Nil K -groups in every degree at most N and every finite number of variables. For $X = \operatorname{Spec} A$, the exponent can also be chosen uniformly over all ind-smooth A -algebras. The proof combines cdh descent and derived Hochschild homology. Products of presheaves pass from finite-dimensional local rings to open neighbourhoods. The corresponding assertion for all Noetherian schemes fails already for one-dimensional domains in degree zero. We give explicit counterexamples in characteristic p and in mixed characteristic; the latter are local and flat over $\mathbb{Z}_{(p)}$ with nonempty regular generic fibre.

1 Introduction

For a quasi-compact quasi-separated scheme X , let $K(X)$ denote the nonconnective algebraic K -theory spectrum of perfect complexes. For an integer $m \geq 1$, set

$$N_m K(X) = \operatorname{fib}\left(K(X \times \mathbb{A}^m) \xrightarrow{0^*} K(X)\right), \quad N_m K_j(X) = \pi_j N_m K(X).$$

Here 0^* evaluates all variables at zero simultaneously. We write $NK = N_1 K$, and use N_m for the same fibre construction on other spectrum-valued functors of schemes. Since zero evaluation has a section, $NK_j(X)$ agrees with $\ker(K_j(\mathbb{A}_X^1) \rightarrow K_j(X))$.

Conjecture A.1 in Morrow's appendix to Chinburg–Pappas–Taylor [6, Appendix A, p. 26] asks whether, whenever X is Noetherian and $X[1/p]$ is regular, each group $NK_j(X)$ is annihilated by a power of p . We prove the assertion under quasi-excellence, in the following uniform form.

Theorem 1.1. *Let p be a prime and let X be a Noetherian quasi-excellent scheme such that $X[1/p]$ is regular. For every integer N , there is an integer $e = e(X, p, N) \geq 0$ such that*

$$p^e N_m K_j(X) = 0 \quad \text{for every } m \geq 1 \text{ and every } j \leq N.$$

The scheme X may be nonseparated and may have infinite Krull dimension. The affine statement has a further uniformity in the coefficient algebra. Recall that an A -algebra is *ind-smooth* if it is a filtered colimit of smooth finitely presented A -algebras, with arbitrary transition homomorphisms.

Theorem 1.2. *Let p be a prime and let A be a Noetherian quasi-excellent commutative ring such that $A[1/p]$ is regular. For every integer N , there is an integer $e = e(A, p, N) \geq 0$ such that*

$$p^e N_m K_j(B) = 0$$

for every ind-smooth A -algebra B , every integer $m \geq 1$, and every integer $j \leq N$.

Finite Zariski Mayer–Vietoris sequences deduce Theorem 1.1 from Theorem 1.2.

The Noetherian hypothesis alone does not give even a bound in a single degree. In Theorem 2.1, for every prime p we construct a one-dimensional Noetherian $\mathbb{F}_p$ -domain R with an injection

$$\bigoplus_{i \geq 1} \mathbb{Z}/p^i \mathbb{Z} \hookrightarrow NK_0(R).$$

Thus Conjecture A.1 as stated in [6, v1] is false. Here $R[1/p] = 0$, and the empty generic fibre is regular. We also give a local mixed-characteristic domain with nonempty regular generic fibre and unbounded p -exponent in NK_0 (Theorem 2.5). These examples fail quasi-excellence. They distinguish bounded exponent in a fixed Nil group from the weaker assertion that each of its elements is killed by some power of p .

Split polynomial evaluation reduces the affine assertion to the single-variable functor NK . Ordinary Nil localization shows that $NK(B)[1/p] = 0$. It remains to prove

$$NK(B)_p^\wedge[1/p] = 0 \tag{1.1}$$

for each ind-smooth algebra B . Together, these two vanishing statements imply bounded p -torsion in each homotopy group by a result of Bhatt on torsion derived-complete groups.

The proof of (1.1) has two separate inputs. First, a cdh-acyclic finite proper geometric relation with regular generic fibres is annihilated by the functor $NK(-)_p^\wedge[1/p]$, even after ind-smooth base change. This is Proposition 4.2. Its proof uses proper coherence for relative derived Hochschild homology and the arithmetic square for topological cyclic homology; its bounds come from finite affine covers. Second, over a finite-dimensional quasi-excellent base, compatible resolution of finite diagrams extends a functor on objects with regular generic fibre to a cdh sheaf. Kelly’s regular alterations and finite-dimensional cdh hypercompleteness then give a vanishing criterion (Proposition 5.1).

Each local ring of A has finite dimension. To obtain a neighbourhood valid for every coefficient algebra, suppose instead that every neighbourhood admits a counterexample to (1.1). Products over successively smaller neighbourhoods combine these counterexamples into a presheaf on the local ring, with a nonzero value on the base. Finite cdh relations descend to neighbourhoods, and quasi-excellence allows their regularity conditions to descend as well. The resulting presheaf therefore satisfies the finite-dimensional vanishing criterion, a contradiction. Sections 6–7 carry out this argument. Finite Zariski descent then proves (1.1) over A .

Finally, a filtered colimit of finite products places any countable collection of the groups $NK_N(B)$ inside one such group as direct summands. Their individual boundedness therefore gives a common exponent. Split polynomial evaluation and the natural nonconnective Bass–Heller–Swan decomposition give the same exponent for all variables and all lower degrees. Finite Mayer–Vietoris induction then preserves this uniformity on a fixed scheme X .

The geometric input is Murayama’s compatible cubical resolution in characteristic zero [16, Theorem 4.3.4 and Proposition 4.4.7]. Appendix A uses it to prove the extension theorem for schemes with regular generic fibre. The quotient comparison there uses the factorization criterion for Verdier quotients [8, Lemma A.5; arXiv Lemma 12.5].

Conventions

All rings are commutative and unital. The finite-type sites used below consist of schemes separated and finitely presented over a Noetherian base. We write $X[1/p] = X \times_{\mathrm{Spec} \mathbb{Z}} \mathrm{Spec} \mathbb{Z}[1/p]$. The empty scheme is regular. For an A -scheme Y and an A -algebra B , write $Y_B = Y \times_{\mathrm{Spec} A} \mathrm{Spec} B$. We write L_{cdh} for cdh sheafification.

For a spectrum E , its derived p -completion is $E_p^\wedge = \varprojlim_r E/p^r$. We call E p -primary if $E[1/p] = 0$. Every prime other than p acts invertibly on $E_p^\wedge$, so $E_p^\wedge[1/p]$ is a rational spectrum. Rational spectra are identified with complexes of $\mathbb{Q}$ -vector spaces [15, Theorem 7.1.2.13]. Complexes are cohomologically graded. The notation Map denotes a mapping spectrum. Hochschild homology is always derived Hochschild homology, and all tensor products of complexes or spectra are derived.

2 A counterexample without quasi-excellence

Theorem 2.1. *For every prime p , there is a one-dimensional Noetherian $\mathbb{F}_p$ -domain R and an injection*

$$\bigoplus_{i \geq 1} \mathbb{Z}/p^i \mathbb{Z} \hookrightarrow NK_0(R).$$

The regular locus of $\text{Spec } R$ consists of its generic point and is not open. In particular, R is not quasi-excellent.

The construction places singularities of increasing order in one Noetherian ring. A separate finite extension at each singularity gives a Milnor square; units in its Artinian quotient then produce independent classes of increasing order in NK_0 .

2.1 The Noetherian ring

Fix $k = \mathbb{F}_p$, put $q_i = p^i$, and take algebraically independent variables $x_1, x_2, \dots$. Define

$$A_i = k + x_i^{q_i} k[x_i] = k[x_i^{q_i}, x_i^{q_i+1}, \dots, x_i^{2q_i-1}], \quad B = \bigotimes_{i \geq 1, k} A_i.$$

The infinite tensor product means the filtered union of the finite tensor products. Each A_i has k -basis $1, x_i^{q_i}, x_i^{q_i+1}, \dots$. Their tensor products identify B with a subring of $k[x_1, x_2, \dots]$; thus B is a domain and each of its elements involves only finitely many variables.

For each i , let

$$C_i = \bigotimes_{j \neq i, k} A_j, \quad \mathfrak{p}_i = (x_i^{q_i}, \dots, x_i^{2q_i-1})B.$$

Then

$$B = C_i + x_i^{q_i} C_i[x_i], \quad \mathfrak{p}_i = x_i^{q_i} C_i[x_i], \quad B/\mathfrak{p}_i = C_i.$$

In particular, each $\mathfrak{p}_i$ is prime. Set

$$S = B \setminus \bigcup_{i \geq 1} \mathfrak{p}_i, \quad R = S^{-1}B, \quad \mathfrak{m}_i = S^{-1}\mathfrak{p}_i.$$

The set S is multiplicatively closed.

Lemma 2.2. *Every ideal $J \subset B$ disjoint from S is contained in some $\mathfrak{p}_i$.*

Proof. The assertion is immediate for $J = 0$. Choose $0 \neq f \in J$. The set $T = \{i : f \in \mathfrak{p}_i\}$ is finite, since setting $x_i = 0$ does not change f when x_i does not occur in it. It is nonempty because $f \notin S$.

Suppose that J is contained in none of the $\mathfrak{p}_i$. For $i \in T$, choose $a_i \in J \setminus \mathfrak{p}_i$, and set

$$g = \sum_{i \in T} a_i \prod_{\ell \in T \setminus \{i\}} x_\ell^{q_\ell}.$$

Modulo $\mathfrak{p}_i$, for $i \in T$, only the i -th term survives, and it is nonzero in the domain C_i . Thus g avoids all these primes. Choose j such that x_j occurs in neither f nor g , and put $h = f + x_j^{q_j} g \in J$. Its image modulo $\mathfrak{p}_j$ is $f \neq 0$. For $i \neq j$, the two terms in its image modulo $\mathfrak{p}_i$ have distinct x_j -degrees. If $i \notin T$, the first term is nonzero; if $i \in T$, the second is nonzero. Hence $h \in S$, a contradiction. $\square$

Lemma 2.3. *The ring R is a one-dimensional Noetherian domain. Its maximal ideals are exactly the $\mathfrak{m}_i$, and each nonzero element belongs to only finitely many of them. If $F_i = \text{Frac}(C_i)$, then*

$$R_{\mathfrak{m}_i} \simeq (F_i + x_i^{q_i} F_i[x_i])_{(x_i^{q_i}, \dots, x_i^{2q_i-1})}. \quad (2.1)$$

Proof. Lemma 2.2 implies that every proper ideal of R is contained in some $\mathfrak{m}_i$. These primes are pairwise incomparable, since $x_i^{q_i}$ belongs to $\mathfrak{p}_i$ and to no $\mathfrak{p}_j$ with $j \neq i$, and each $\mathfrak{p}_i$ is disjoint from S . Consequently the $\mathfrak{m}_i$ are precisely the maximal ideals. For $0 \neq f/s \in R$, membership in $\mathfrak{m}_i$ is equivalent to $f \in \mathfrak{p}_i$, which occurs for only finitely many i .

Since $C_i \setminus \{0\}$ is disjoint from $\mathfrak{p}_i$, localizing gives (2.1). The ring on the right is the localization of a one-dimensional finitely generated domain over F_i , and hence is Noetherian of dimension one.

To prove that R itself is Noetherian, let $I \neq 0$ be an ideal and choose $0 \neq f \in I$. Only finitely many maximal ideals contain f . At each of them, choose finitely many elements of I whose images generate the localized ideal. Let I_0 be the ideal generated by all these elements together with f . Then $(I/I_0)_{\mathfrak{m}} = 0$ at every maximal ideal: this follows from the choice of generators where $f \in \mathfrak{m}$, and from the invertibility of f elsewhere. Localization at all maximal ideals detects the zero module, so $I = I_0$. Thus every ideal is finitely generated. Formula (2.1) also gives $\dim R = 1$. $\square$

2.2 Milnor squares and Nil classes

For each i , the element

$$x_i = \frac{x_i^{q_i+1}}{x_i^{q_i}}$$

belongs to $\text{Frac}(R)$ and is integral over R . Thus $E_i = R[x_i]$ is a finite extension of domains. The description of $\mathfrak{p}_i$ above gives

$$\mathfrak{m}_i = x_i^{q_i} E_i,$$

so $\mathfrak{m}_i$ is a common ideal of R and E_i . For $0 \neq c \in C_i$, the element $c + x_i^{q_i}$ belongs to S : its image modulo $\mathfrak{p}_i$ is c , and modulo any other $\mathfrak{p}_j$ the term $x_i^{q_i}$ cannot cancel with a term independent of x_i . Inverting these elements gives

$$R/\mathfrak{m}_i = F_i, \quad E_i/\mathfrak{m}_i = D_i := F_i[\epsilon]/(\epsilon^{q_i}),$$

where x_i maps to ϵ . Consequently the square

$$\begin{array}{ccc} R & \longrightarrow & E_i \\ \downarrow & & \downarrow \\ F_i & \longrightarrow & D_i \end{array} \quad (2.2)$$

is cartesian, and its right vertical map is surjective. Indeed, the preimage of $F_i \subset D_i$ is R , since the common ideal is already contained in R . Polynomial extension preserves this square.

For a unit $u \in D_i[t]^\times$, Milnor patching gives a line bundle

$$L(u) = \{(a, b) \in E_i[t] \oplus F_i[t] : \bar{a} = ub\}$$

over $R[t]$, trivialized on each of the two rings used for patching. The associated boundary homomorphism is

$$D_i[t]^\times \longrightarrow K_0(R[t]), \quad u \longmapsto [L(u)] - [R[t]]; \quad (2.3)$$

see [20, I.3.10 and III.2.6]. Its additivity can also be read directly from the patching matrices:

$$\text{diag}(u, v) = \text{diag}(uv, 1) \text{diag}(v^{-1}, v).$$

The last factor is a product of elementary matrices, each of which lifts along the surjection $E_i[t] \rightarrow D_i[t]$. Hence

$$L(u) \oplus L(v) \simeq L(uv) \oplus R[t],$$

which proves that (2.3) converts multiplication into addition.

Take

$$u_i = 1 + \epsilon t, \quad L_i = L(u_i), \quad \alpha_i = [L_i] - [R[t]].$$

The element u_i is a unit since ϵ is nilpotent, and its evaluation at $t = 0$ is 1. Thus $L_i|_{t=0} \simeq R$ and $\alpha_i \in NK_0(R)$.

Lemma 2.4. *The class α_i has additive order p^i in $NK_0(R)$. Its image in $NK_0(R_{\mathfrak{m}_i})$ also has order p^i , whereas its image in $NK_0(R_{\mathfrak{m}_j})$ is zero for $j \neq i$.*

Proof. In characteristic p one has

$$u_i^{p^r} = 1 + \epsilon^{p^r} t^{p^r}.$$

This equals 1 for $r = i$, and is nonconstant in t for $0 \leq r < i$. Since E_i and F_i are domains, their polynomial units are constant. A trivialization of $L(u_i^{p^r})$ would give units $a \in E_i[t]^\times$ and $b \in F_i[t]^\times$ with $u_i^{p^r} = \bar{a}b^{-1}$. The right side is constant, so such a trivialization cannot exist for $r < i$. Patching respects tensor products, and therefore L_i has exact tensor order p^i .

The homomorphism (2.3) gives $p^i \alpha_i = 0$. Conversely, the determinant homomorphism from $K_0(R[t])$ to $\text{Pic}(R[t])$ sends α_i to L_i . The exact tensor order just proved implies that α_i has additive order p^i .

If $j \neq i$, then $x_i^{q_i} \notin \mathfrak{m}_j$, so $x_i \in R_{\mathfrak{m}_j}$ and $E_i \otimes_R R_{\mathfrak{m}_j} = R_{\mathfrak{m}_j}$. The trivialization of L_i over $E_i[t]$ thus makes its image over $R_{\mathfrak{m}_j}[t]$ trivial. At $\mathfrak{m}_i$, the lower rings of (2.2) remain F_i and D_i : every element of $R \setminus \mathfrak{m}_i$ maps to a nonzero constant in $F_i \subset D_i$, hence to a unit. The upper right ring remains a domain. The same unit and determinant arguments therefore show that the localized class still has order p^i . $\square$

Proof of Theorem 2.1. The classes α_i define a homomorphism

$$\bigoplus_{i \geq 1} \mathbb{Z}/p^i \mathbb{Z} \longrightarrow NK_0(R).$$

If a finite relation $\sum_i a_i \alpha_i = 0$ holds, localize at $\mathfrak{m}_j$. Lemma 2.4 leaves only the term with index j , and implies $p^j \mid a_j$. This holds for every j , proving injectivity.

In the local ring (2.1), the monomials

$$x_i^{q_i}, x_i^{q_i+1}, \dots, x_i^{2q_i-1}$$

form an F_i -basis of the maximal ideal modulo its square. Its embedding dimension is therefore $q_i > 1$, whereas its dimension is one. Every closed point of $\text{Spec } R$ is singular; the generic point is regular. Each nonempty principal open subset contains all but finitely many of the $\mathfrak{m}_i$, by Lemma 2.3. Hence the regular locus, consisting of the generic point alone, is not open. The ring fails the J -1 condition and in particular is not quasi-excellent. $\square$

2.3 A local counterexample in mixed characteristic

The failure also occurs for a flat local domain with nonempty regular generic fibre. The following example permits an explicit computation in degree zero.

Theorem 2.5. *For every prime p , there is a one-dimensional Noetherian local domain R , flat over $\mathbb{Z}_{(p)}$, such that $R[1/p]$ is a field and*

$$NK_0(R) \cong \bigoplus_{r \geq 1} \mathbb{Q}_p / \mathbb{Z}_p.$$

In particular, no power of p annihilates $NK_0(R)$.

Proof. We use Akizuki's construction in the form given by Reid [17]. Put

$$n_i = 2((i+1)! - 1), \quad z = \sum_{i \geq 0} p^{n_i} \in \mathbb{Z}_p, \quad z_i = \frac{z - \sum_{j < i} p^{n_j}}{p^{n_i}}, \quad b_i = z_i - 1,$$

and define

$$C = \mathbb{Z}_{(p)}[p(z-1), b_i^2 \mid i \geq 0], \quad R = C_{(p, p(z-1))}. \quad (2.4)$$

The exponents satisfy $n_{i+1} \geq 2n_i + 2$, and z is transcendental over $\mathbb{Q}$. For the latter assertion, let $s_N = \sum_{i \leq N} p^{n_i}$. If $0 \neq f \in \mathbb{Z}[T]$ and $f(z) = 0$, then, for all sufficiently large N , the nonzero integer $f(s_N)$ is divisible by $p^{n_{N+1}}$. On the other hand,

$$|f(s_N)| \leq c_f p^{(\deg f) n_N}.$$

This contradicts $n_{N+1}/n_N \rightarrow \infty$. Reid's theorem therefore applies: R is a one-dimensional Noetherian local domain with fraction field $\mathbb{Q}(z)$. Since p is a nonzero element of its maximal ideal, $R[1/p] = \mathbb{Q}(z)$. The ring R is torsion-free, hence flat, over the discrete valuation ring $\mathbb{Z}_{(p)}$.

We next identify its completion, keeping track of the coefficient ring. Write $w = p(z-1) \in R$, and let $\alpha = p(z-1) \in \mathbb{Z}_p$ denote the corresponding scalar. Huang–Lin's normal form [11, Lemmas 1–2 and Corollary 7] says that $R/p^N R$ is free over $\mathbb{Z}/p^N$, with basis $1, w$. Consequently the p -adic completion $\hat{R}$ is free over $\mathbb{Z}_p$, with that basis. The p -adic and maximal-ideal-adic topologies agree because p is a parameter. For $\alpha_i = p \sum_{j=1}^i p^{n_j}$, the identities in R give

$$(w - \alpha_i)^2 = p^{2n_i+2} b_i^2.$$

Passing to the completion shows that $(w - \alpha)^2 = 0$. The change of basis from $1, w$ to $1, w - \alpha$ is invertible, so

$$\hat{R} \cong \mathbb{Z}_p[\epsilon]/(\epsilon^2), \quad w \mapsto \alpha + \epsilon. \quad (2.5)$$

The completion homomorphism $R \rightarrow \hat{R}$ is an analytic isomorphism along the powers of p : it preserves the quotients modulo p^N , and multiplication by p^N is injective on both rings.

The same assertions hold for the ordinary polynomial extension $R[t] \rightarrow \widehat{R}[t]$. Analytic excision [13, Example 2.9(ii)], followed by the fibre of evaluation at $t = 0$, gives

$$NK(R) \simeq \text{fib}\left(NK(\widehat{R}) \longrightarrow NK(\widehat{R}[1/p])\right),$$

because $R[1/p]$ is a field.

Nilpotent invariance of K_0 and regularity of $\mathbb{Z}_p$ and $\mathbb{Q}_p$ give $NK_0(\widehat{R}) = NK_0(\widehat{R}[1/p]) = 0$. For a split square-zero extension of a commutative ring E , the determinant identifies the relative K_1 -group of $E[\epsilon]/(\epsilon^2) \rightarrow E$ with $1 + \epsilon E$; see [20, III, Lemma 2.4]. Applying this to $E = \mathbb{Z}_p[t]$ and $E = \mathbb{Q}_p[t]$, and using homotopy invariance for the coefficient rings, yields

$$NK_1(\widehat{R}) = 1 + \epsilon t \mathbb{Z}_p[t], \quad NK_1(\widehat{R}[1/p]) = 1 + \epsilon t \mathbb{Q}_p[t].$$

Under multiplication these groups identify with the corresponding additive polynomial groups. The long exact sequence of the displayed fibre therefore gives

$$\begin{aligned} NK_0(R) &\cong \text{coker}\left(NK_1(\widehat{R}) \longrightarrow NK_1(\widehat{R}[1/p])\right) \\ &\cong t(\mathbb{Q}_p/\mathbb{Z}_p)[t] \cong \bigoplus_{r \geq 1} \mathbb{Q}_p/\mathbb{Z}_p. \end{aligned}$$

Each summand has elements of order p^N for every $N \geq 1$. □

This ring is not quasi-excellent. Indeed, its generic formal fibre is

$$\widehat{R} \otimes_R \mathbb{Q}(z) = \widehat{R}[1/p] \cong \mathbb{Q}_p[\epsilon]/(\epsilon^2),$$

which is not regular.

3 Nil localization and completion

We use nonconnective K -theory with its Zariski and Nisnevich descent, filtered-colimit continuity on rings, and polynomial homotopy invariance on regular Noetherian schemes [19, Sections 7–9]. These statements also give continuity for schemes obtained by affine base change, using finite affine covers and finite Mayer–Vietoris squares.

Lemma 3.1. *For every quasi-compact quasi-separated scheme X and $m \geq 1$, there is a natural equivalence*

$$N_m K(X) \simeq \bigoplus_{a=0}^{m-1} NK(X \times \mathbb{A}^a).$$

Proof. Evaluation of the last variable at zero is split by the projection, so

$$K(X \times \mathbb{A}^{a+1}) \simeq K(X \times \mathbb{A}^a) \oplus NK(X \times \mathbb{A}^a).$$

Iterate these splittings and take the fibre of simultaneous zero evaluation to obtain the displayed formula. □

Lemma 3.2. *Let A be a Noetherian ring such that $A[1/p]$ is regular, and let B be an ind-smooth A -algebra. Then*

$$N_m K(B)[1/p] = 0 \quad (m \geq 1).$$

Moreover, the natural map

$$N_m K(B) \longrightarrow N_m K(B \otimes_{\mathbb{Z}} \mathbb{Z}_{(p)})$$

is an equivalence. The first assertion also holds for $N_m K(Y_B)$, where Y is a Noetherian quasi-compact quasi-separated A -scheme with $Y[1/p]$ regular.

Proof. We use the integer-localization formula for Nil groups: for a commutative ring R , a multiplicative set of integers S , and $j \in \mathbb{Z}$,

$$S^{-1}NK_j(R) \cong NK_j(R \otimes_{\mathbb{Z}} S^{-1}\mathbb{Z}). \quad (3.1)$$

For arbitrary commutative rings, the formula is recorded in the proof of [5, Lemma 4.61 and footnote 16]; see also [19, Exercise 9.12]. Lemma 3.1 gives the same localization formula for N_mK_j .

Write $B = \operatorname{colim}_{\lambda} B_{\lambda}$, with each B_{λ} smooth and finitely presented over A . Every $B_{\lambda}[1/p]$ is regular. Homotopy invariance, (3.1), and continuity of ordinary K -theory therefore give $N_mK(B)[1/p] = 0$. The same argument on affine opens of Y , followed by finite Zariski gluing, proves the assertion for Y_B .

Each homotopy group of $N_mK(B)$ is consequently p -primary. Localizing such a group at all integers prime to p leaves it unchanged. Apply (3.1) once more to obtain the asserted equivalence of spectra. $\square$

Lemma 3.3. *Let E be a spectrum with $E[1/p] = 0$. The following conditions are equivalent:*

- (i) *for each $j \in \mathbb{Z}$, the group $\pi_j E$ is annihilated by some power of p ;*
- (ii) *E is derived p -complete;*
- (iii) *$E_p^{\wedge}[1/p] = 0$.*

Proof. Derived p -completeness is equivalent to the vanishing of the inverse limit of the tower $\cdots \xrightarrow{p} E \xrightarrow{p} E$. If (i) holds, each tower of homotopy groups is pro-zero, so the Milnor exact sequence proves (ii). Conversely, if this inverse limit of spectra vanishes, the same exact sequence gives

$$\varprojlim(\pi_j E, p) = 0, \quad \varprojlim^1(\pi_j E, p) = 0 \quad \text{for every } j.$$

Thus each group $\pi_j E$, regarded as a complex in degree zero, is derived p -complete. It is also p -primary by hypothesis. Bhatt's theorem on torsion derived completions [2, Theorem 0.1] says that such a group has bounded p -exponent. This proves (i).

Finally, E is p -local, and its arithmetic fracture square is

$$\begin{array}{ccc} E & \longrightarrow & E[1/p] \\ \downarrow & & \downarrow \\ E_p^{\wedge} & \longrightarrow & E_p^{\wedge}[1/p]. \end{array}$$

Since the upper right corner vanishes, (ii) and (iii) are equivalent. $\square$

Lemma 3.4. *Let A be Noetherian, let B be ind-smooth over A , and fix $m \geq 1$.*

- (i) *If T is a regular finitely presented A -scheme, then $N_mK(T_B) = 0$.*
- (ii) *If $f : Y \rightarrow T$ is finite locally free with a specified isomorphism $f_*\mathcal{O}_Y \simeq \mathcal{O}_T^{\oplus d}$, then pullback and pushforward induce maps on Nil spectra with*

$$f_*f^* \simeq d \operatorname{id}.$$

The same homotopy holds after derived completion and inversion of p .

Proof. For (i), the schemes T_{B_λ} are regular for a smooth presentation $B = \operatorname{colim}_\lambda B_\lambda$. Apply homotopy invariance and ordinary K -theory continuity. For (ii), finite locally free pushforward preserves perfect complexes. The projection formula and the chosen basis identify

$$f_* f^* P \simeq P \otimes f_* \mathcal{O}_Y \simeq P^{\oplus d}.$$

Additivity of K -theory gives the homotopy. This identification commutes with polynomial extension and zero evaluation, so it gives the asserted homotopy on their fibres. $\square$

4 Finite proper relations

We prove that finite proper cdh descent diagrams with regular generic fibres are annihilated by $N_m K(-)_p^\wedge[1/p]$.

Definition 4.1. A *finite geometric relation* consists of a finite diagram of schemes and a specified finite homotopy construction on its contravariant values: a cubical total fibre, an iterated total fibre, or the fibre of a geometric comparison between such constructions. We write $\delta F(D)$ for the resulting spectrum. We call the relation *cdh-acyclic* if

$$\delta F(D) = 0$$

for every spectrum-valued cdh sheaf F . The relation is *proper over T* if all its vertices and arrows are over T , and every vertex is proper over T .

For example, a commutative square

$$\begin{array}{ccc} W & \longrightarrow & V \\ \downarrow & & \downarrow \\ U & \longrightarrow & T \end{array}$$

defines the total fibre

$$\delta F = \operatorname{fib}\left(F(T) \longrightarrow F(U) \times_{F(W)} F(V)\right). \quad (4.1)$$

For a Nisnevich distinguished square or an abstract blow-up square, this fibre vanishes for every cdh sheaf. The construction specified in Definition 4.1 records which finite fibres are taken and the maps between them. It is exact. Moreover, cdh-acyclicity is preserved by base change: precomposition with base change takes a cdh sheaf to a cdh sheaf, since base change preserves Nisnevich and abstract blowup squares.

Proposition 4.2. *Let A be a Noetherian $\mathbb{Z}_{(p)}$ -algebra, let T be a separated scheme of finite type over A , and let D be a cdh-acyclic finite geometric relation proper over T . Suppose that $Y[1/p]$ is regular for every vertex Y of D . For every ind-smooth A -algebra B and every $m \geq 1$, the spectrum*

$$\delta N_m K(D_B)$$

is derived p -complete and becomes zero after inverting p . Consequently its homotopy groups have bounded p -power exponent in each degree, and

$$\delta(N_m K(D_B)_p^\wedge[1/p]) = 0.$$

The proof has two Hochschild-homology inputs: cdh descent on regular characteristic-zero schemes and degreewise torsion bounds for proper diagrams.

Lemma 4.3. *For a regular Noetherian $\mathbb{Q}$ -scheme Y , the natural map*

$$HH(Y/\mathbb{Q}) \longrightarrow (L_{\text{cdh}} HH(-/\mathbb{Q}))(Y)$$

is an equivalence, compatible with the circle action. The same holds for a scheme obtained from such a Y by an ind-smooth affine base change.

Proof. For smooth schemes of finite type over $\mathbb{Q}$, this is the Hochschild comparison of Cortiñas–Haesemeyer–Weibel [7, Theorem 2.4]. If R is a regular Noetherian $\mathbb{Q}$ -algebra, then $\mathbb{Q} \rightarrow R$ is geometrically regular. Popescu’s theorem therefore writes R as a filtered colimit of smooth finitely presented $\mathbb{Q}$ -algebras [18, Tag 07GB]. Hochschild homology is finitary, and cdh sheafification with values in a presentable stable category preserves finitary presheaves [1, Proposition 2.15(2)]. Taking this colimit proves the comparison for $\text{Spec } R$. Zariski descent then proves it for Y .

Hochschild homology and its comparison with cdh sheafification take values in $\text{Fun}(BS^1, \text{Sp})$, so this equivalence is compatible with the circle action.

For the last assertion, write the base change as a filtered colimit of smooth finitely presented base changes. Each resulting Noetherian scheme is regular. The finitary property on both sides, followed by finite Zariski gluing, gives the assertion. $\square$

Lemma 4.4. *Let H be a connective ring spectrum, put $R = \pi_0 H$, and let M be an H -module bounded below. If each homotopy group of $R \otimes_H M$ is annihilated by a power of p , then the same is true of each homotopy group of M .*

Proof. Choose b such that M is b -connective, and let $J = \text{fib}(H \rightarrow R)$. This is a 1-connective H -bimodule, and there is a fibre sequence

$$J \otimes_H M \longrightarrow M \longrightarrow R \otimes_H M.$$

Suppose that p^{c_n} annihilates $\pi_n(R \otimes_H M)$. We construct exponents a_n for $\pi_n M$, starting with $a_b = c_b$. For $n > b$, connectivity gives

$$\pi_n(J \otimes_H M) \simeq \pi_n(J \otimes_H \tau_{\leq n-1} M).$$

The finite Postnikov tower of $\tau_{\leq n-1} M$ has layers $\pi_j M[j]$, for $b \leq j < n$. Each such layer is annihilated, as an H -module, by p^{a_j} . If the outer terms of a fibre sequence are annihilated by p^a and $p^{a'}$, its middle term is annihilated by $p^{a+a'}$; this follows by factoring the appropriate multiplication map through the first term. Hence $\tau_{\leq n-1} M$, and then its tensor product with J , are annihilated by $p^{\sum_{j=b}^{n-1} a_j}$. The displayed fibre sequence now permits the choice

$$a_n = c_n + \sum_{j=b}^{n-1} a_j.$$

This completes the induction. $\square$

Lemma 4.5. *Let R be a Noetherian $\mathbb{Z}_{(p)}$ -algebra, and let D be a finite geometric relation proper over $\text{Spec } R$. If*

$$\delta HH(D/\mathbb{Z})[1/p] = 0,$$

then, for every flat R -algebra C ,

$$(\delta HH(D_C/\mathbb{Z}))_p^\wedge[1/p] = 0.$$

Proof. The circle-tensor description of derived Hochschild homology gives, for every R -algebra R' ,

$$R \otimes_{HH(R/\mathbb{Z})} HH(R'/\mathbb{Z}) \simeq HH(R'/R);$$

see [3, §2.2, Remarks 2.3–2.4]. Each vertex of D has a finite affine open cover with affine intersections. Zariski descent for these covers can be expressed by finite Mayer–Vietoris constructions. Since tensor product is exact, the affine equivalence glues and commutes with δ , giving

$$R \otimes_{HH(R/\mathbb{Z})} \delta HH(D/\mathbb{Z}) \simeq \delta HH(D/R). \quad (4.2)$$

We claim that $\delta HH(D/R)$ is bounded below and that each of its homotopy groups is a finite R -module. For a vertex Y , the relative cotangent complex $L_{Y/R}$ is connective and pseudocoherent, as are its fixed derived exterior powers. The derived HKR filtration has graded pieces

$$\bigwedge^w L_{Y/R}[w].$$

Choose finite affine covers of the vertices, and let d bound their alternating Čech widths. Choose s bounding the loss of connectivity caused by the finite construction δ . The contribution of weight w is at least $(w - d - s)$ -connective. For fixed n , the filtration tail in sufficiently high weights is therefore $(n + 1)$ -connective and does not affect π_n . The remaining weights require only finite Postnikov truncations. Proper coherent cohomology is finite over R [18, Tag 02O3], proving the claim. The same finite covers, applied to absolute Hochschild homology, also give a lower bound for $\delta HH(D/\mathbb{Z})$.

By hypothesis and (4.2), $\delta HH(D/R)[1/p] = 0$. Its finite homotopy modules are therefore each annihilated by some power of p . Flat base change, again computed on the same finite covers, gives

$$\begin{aligned} \delta HH(D_C/C) &\simeq \delta HH(D/R) \otimes_R C, \\ \pi_n \delta HH(D_C/C) &= \pi_n \delta HH(D/R) \otimes_R C. \end{aligned}$$

In particular, these homotopy groups inherit the same exponents. The affine base-change formula and finite gluing give

$$C \otimes_{HH(C/\mathbb{Z})} \delta HH(D_C/\mathbb{Z}) \simeq \delta HH(D_C/C).$$

Here $HH(C/\mathbb{Z})$ is connective with zeroth homotopy ring C , and $\delta HH(D_C/\mathbb{Z})$ has the lower bound furnished by the pulled-back finite covers and δ . Lemma 4.4 therefore shows that every homotopy group of $\delta HH(D_C/\mathbb{Z})$ is annihilated by a power of p . The completion criterion of Lemma 3.3 gives the asserted vanishing. $\square$

Proof of Proposition 4.2. By Lemma 3.1 and stability of ind-smoothness under polynomial extension, it suffices to treat $m = 1$. First suppose $T = \operatorname{Spec} R$, and put $C = R \otimes_A B$. This is ind-smooth, and hence flat, over R . For each smooth finitely presented stage B_λ of B , the schemes $(Y_{B_\lambda})[1/p]$ are regular Noetherian $\mathbb{Q}$ -schemes. Lemma 4.3 and its finitary extension therefore identify Hochschild homology with its cdh sheafification at every vertex of $(D_B)[1/p]$. Since the relation is cdh-acyclic, this gives

$$\delta HH((D_B)[1/p]/\mathbb{Q}) = 0 \quad (4.3)$$

as a spectrum with circle action. Taking homotopy fixed points, which commute with finite limits, gives

$$\delta HC^-((D_B)[1/p]/\mathbb{Q}) = 0. \quad (4.4)$$

The same argument with $B = A$ proves the hypothesis of Lemma 4.5. That lemma and flatness of C yield

$$(\delta HH(D_B/\mathbb{Z}))_p^\wedge[1/p] = 0. \quad (4.5)$$

All these statements also apply to the polynomial diagram $D_B \times \mathbb{A}^1$. For this application its proper Noetherian base is $\text{Spec } R[t]$. Consequently (4.4) and (4.5) remain valid after taking the fibre of zero evaluation.

For every qcqs $\mathbb{Z}_{(p)}$ -scheme Z there is a Cartesian square

$$\begin{array}{ccc} TC(Z) & \longrightarrow & HC^-(Z[1/p]/\mathbb{Q}) \\ \downarrow & & \downarrow \\ TC(Z)_p^\wedge & \longrightarrow & (HH(Z/\mathbb{Z})_p^\wedge[1/p])^{hS^1}. \end{array} \quad (4.6)$$

Indeed, the unfiltered TC-negative-cyclic square of [5, §2] reduces over $\mathbb{Z}_{(p)}$ to

$$\begin{array}{ccc} TC(Z) & \longrightarrow & HC^-(Z/\mathbb{Z}) \\ \downarrow & & \downarrow \\ TC(Z)_p^\wedge & \longrightarrow & HC^-(Z/\mathbb{Z})_p^\wedge. \end{array}$$

Paste this with the square obtained by applying circle homotopy fixed points to the arithmetic square for $HH(Z/\mathbb{Z})$; see also [5, Equation (4.15.1) and Remark 4.17]. Completion commutes with homotopy fixed points, since both are limits and reduction modulo p^r is an exact finite construction. This proves (4.6).

The presheaf

$$K^{\text{inv}} = \text{fib}(K \longrightarrow TC)$$

is a cdh sheaf on qcqs schemes [13, Theorem E and Corollary A.4]. Its base-change and polynomial presheaves are also cdh sheaves. Hence cdh-acyclicity gives an equivalence

$$\delta NK(D_B) \simeq \delta N_1 TC(D_B).$$

Apply the finite construction δN_1 to (4.6). The upper right corner vanishes by (4.4), and the lower right corner vanishes by (4.5); completion, rationalization, and circle homotopy fixed points commute with the finite construction. The left vertical map is therefore an equivalence. This proves that $\delta NK(D_B)$ is derived p -complete.

Lemma 3.2 gives

$$NK(Y_B)[1/p] = 0$$

for each vertex Y , and hence $\delta NK(D_B)[1/p] = 0$. Lemma 3.3 proves both the degreewise exponent bound and the final vanishing in the proposition.

For general T , choose a finite affine open cover. The relation restricts to one satisfying the affine hypotheses on each intersection. The functors involved satisfy Zariski descent, and finite Mayer–Vietoris gluing preserves derived p -completeness and vanishing after inverting p . The affine conclusion therefore glues to T . $\square$

5 Vanishing over a finite-dimensional base

Let S be a finite-dimensional Noetherian quasi-excellent $\mathbb{Z}_{(p)}$ -scheme with $S[1/p]$ regular. Write $\mathcal{D}$ for the category of separated finitely presented S -schemes, and $\mathcal{C} \subset \mathcal{D}$ for the full subcategory of schemes with regular generic fibre. The category $\mathcal{D}$ includes nonreduced schemes and has the usual fibre products.

Let $\mathcal{R}_{\mathcal{C}}$ be the following collection of geometric relations: the empty-scheme and finite-disjoint-union relations, Nisnevich distinguished squares in $\mathcal{C}$, and the cdh-acyclic finite proper relations of Definition 4.1 with vertices in $\mathcal{C}$. The common proper base of the last kind of relation can be any object of $\mathcal{D}$. A presheaf annihilates a relation when the associated total fibre is zero.

Proposition 5.1. *Let $G : \mathcal{D}^{\text{op}} \rightarrow \text{Sp}_{\mathbb{Q}}$ be a rational spectrum-valued presheaf. Suppose that:*

- (i) $G(T) = 0$ for every regular $T \in \mathcal{D}$;
- (ii) $G|_{\mathcal{C}}$ annihilates $\mathcal{R}_{\mathcal{C}}$;
- (iii) for every finite flat surjection $f : Y \rightarrow T$ with $f_*\mathcal{O}_Y$ globally free of positive constant rank, the map $\pi_j G(T) \rightarrow \pi_j G(Y)$ is injective for every j .

Then $G(S) = 0$.

The proof combines Kelly's regular alterations with the extension theorem in Appendix A. The alterations kill the cdh homotopy sheaves of G . The extension theorem supplies a cdh sheaf agreeing with G on $\mathcal{C}$, and finite-dimensional hypercompleteness then kills its values.

Proof. Choose a prime $\ell \neq p$, which is invertible on S . An $\text{fps}\ell'$ -cover is a finite flat surjection whose direct image is globally free of rank prime to ℓ . By (iii), every presheaf $\pi_j G$ is separated for these covers. Kelly's refinability and separatedness results [12, Example 3.4.6 and Proposition 3.4.8(1)] give an injection

$$a_{\text{cdh}}\pi_j G \longrightarrow a_{\ell\text{dh}}\pi_j G.$$

Every object of $\mathcal{D}$ admits an ℓdh -cover by regular schemes [12, Corollary 3.2.13]. By (i), any section of $\pi_j G$ vanishes on such a cover. Since sections of a sheafification are locally represented by presheaf sections,

$$a_{\text{cdh}}\pi_j G = 0 \quad (j \in \mathbb{Z}). \tag{5.1}$$

Apply Theorem A.1 to (ii). It gives a rational cdh sheaf H with a natural equivalence $H|_{\mathcal{C}} \simeq G|_{\mathcal{C}}$, and shows that $\mathcal{C}$ is covering-dense. Given a section of $\pi_j H(T)$, pull it back to a cdh cover by objects of $\mathcal{C}$. It then corresponds to a section of $\pi_j G$, which vanishes on a further cdh cover by (5.1). Refine this cover again by objects of $\mathcal{C}$. The natural equivalence with G shows that the original H -section is locally zero. Thus $a_{\text{cdh}}\pi_j H = 0$ for all j .

For a Noetherian scheme, valuative dimension equals Krull dimension [9, Proposition 2.3.2(9)]. The finite-presentation cdh topos over S is therefore hypercomplete [9, Corollary 2.4.16]. For every $a \in \mathbb{Z}$, the space-valued sheaf $\Omega^{\infty}\Sigma^a H$ has trivial homotopy sheaves and is consequently contractible. Hence $H = 0$. Since $S \in \mathcal{C}$, we obtain $G(S) \simeq H(S) = 0$. $\square$

Corollary 5.2. *Let A be a finite-dimensional Noetherian quasi-excellent $\mathbb{Z}_{(p)}$ -algebra with $A[1/p]$ regular. For every ind-smooth A -algebra B and every $m \geq 1$,*

$$N_m K(B)_p^{\wedge}[1/p] = 0.$$

Proof. Fix B . The rational spectrum-valued presheaf

$$T \longmapsto NK(T_B)_p^{\wedge}[1/p]$$

vanishes on regular objects and has injective finite-free pullbacks by Lemma 3.4. It satisfies Nisnevich descent and annihilates the proper relations by Proposition 4.2. Proposition 5.1 therefore makes its value on $\text{Spec } A$ zero. Apply this to the polynomial extensions of B , and use Lemma 3.1 for the assertion with m variables. $\square$

6 Descending finite cdh relations

We will apply Proposition 5.1 over local rings. The passage to an open neighbourhood uses the following finiteness property of cdh descent. Throughout this section, cdh sheaves take values in spectra with integral coefficients.

For a Noetherian scheme S , let $\mathcal{D}_S$ denote its category of separated finitely presented schemes. The spectral representable of $T \in \mathcal{D}_S$ is

$$\mathbb{S}[T] = \Sigma_+^\infty \operatorname{Hom}_{\mathcal{D}_S}(-, T).$$

A finite geometric relation D determines a finite-cell presheaf $\operatorname{Rel}(D)$, obtained by applying its prescribed finite cofibres to spectral representables. Thus

$$\operatorname{Map}(\operatorname{Rel}(D), F) \simeq \delta F(D).$$

For the square in (4.1), this is the cofiber of $\mathbb{S}[U] \amalg_{\mathbb{S}[W]} \mathbb{S}[V] \rightarrow \mathbb{S}[T]$. Here finite-cell means obtained from finitely many representables by shifts and finite cofibres.

Lemma 6.1. *Let A be Noetherian and quasi-excellent, and let $\mathfrak{q} \subset A$ be a prime ideal. A finite diagram of separated finitely presented $A_{\mathfrak{q}}$ -schemes descends to A_f for some $f \notin \mathfrak{q}$. One can preserve designated proper morphisms, Nisnevich squares, finite free morphisms with chosen bases, regular vertices, vertices with regular generic fibre, and the cdh acyclicity of finitely many finite geometric relations in the diagram.*

Proof. First work over any Noetherian base S . Let $\mathcal{E}$ consist of $\mathbb{S}[\emptyset]$ and the relation objects of Nisnevich and abstract blow-up squares. The finite-square criterion identifies cdh sheaves with the presheaves right orthogonal to $\mathcal{E}$ [9, Propositions 2.1.5–2.1.6]. All these objects are compact. Write $\operatorname{thick}(\mathcal{E})$ for their closure under finite sums, shifts, cofibres, and retracts. A compact presheaf annihilated by cdh sheafification belongs to this closure. Indeed, the inclusion of $\operatorname{thick}(\mathcal{E})$ extends to a fully faithful functor from its Ind-category, formed by adjoining filtered colimits. Its image is the localizing subcategory generated by $\mathcal{E}$, where arbitrary colimits are allowed, and is the kernel of cdh sheafification. The compact objects of that Ind-category are precisely $\operatorname{thick}(\mathcal{E})$; see [4, §2.4, Lemma 2.20, and Proposition 5.6]. Applied to $\operatorname{Rel}(D)$, this expresses it as a retract of a finite iterated cofiber of shifts of finitely many objects of $\mathcal{E}$.

Apply this observation over $A_{\mathfrak{q}}$. Finite-presentation limit theorems descend the geometric diagram and the finitely many square relations to some A_f [18, Tag 01ZM]. The squares remain Nisnevich or abstract blow-up squares after further localization, as in the proof of [9, Proposition 2.1.6]. The finite-cell expression and its retraction descend as well. Indeed, between spectral representables the mapping spectrum is

$$\operatorname{Map}(\mathbb{S}[U], \mathbb{S}[V]) \simeq \Sigma_+^\infty \operatorname{Hom}(U, V).$$

Descent of morphisms identifies its value over $A_{\mathfrak{q}}$ with the filtered colimit of its values over the neighbourhoods. Induction on finite cofibres gives the same assertion for maps between finite-cell objects, since filtered colimits of spectra commute with finite limits. Thus the attaching maps and both retraction maps descend; the equality of their composite with the identity holds after a further localization, by the same assertion on π_0 . The descended relation therefore still belongs to the thick closure of the cdh square relations and is cdh-acyclic.

Proper morphisms descend by [18, Tag 081F]. Finite free morphisms descend with their bases and inverse basis isomorphisms. For regularity, a descended finitely presented

scheme T has closed singular locus because A is quasi-excellent. If $T_{A_{\mathfrak{q}}}$ is regular, that closed locus becomes empty after base change to $A_{\mathfrak{q}}$. Give it a closed subscheme structure; its emptiness descends, using a finite affine cover, to some neighbourhood. This makes T regular there. Applying the same argument to $T[1/p]$ preserves regular generic fibres. A common localization handles the finitely many conditions. $\square$

7 A uniform neighbourhood

Proposition 7.1. *Let A be a Noetherian quasi-excellent $\mathbb{Z}_{(p)}$ -algebra with $A[1/p]$ regular. For every prime ideal $\mathfrak{q} \subset A$, there is $f \in A \setminus \mathfrak{q}$ such that*

$$N_m K(B)_p^\wedge[1/p] = 0$$

for every ind-smooth A_f -algebra B and every $m \geq 1$.

The local ring $A_{\mathfrak{q}}$ has finite dimension, so Proposition 5.1 applies there. If no neighbourhood had the asserted uniformity, we could combine its counterexamples into a presheaf over $A_{\mathfrak{q}}$. This presheaf would satisfy that proposition's hypotheses but have a nonzero value on the base.

Proof. By Lemma 3.1, it suffices to prove the assertion for NK : polynomial extensions of ind-smooth algebras remain ind-smooth. Let $\mathcal{U}$ be the set of principal open neighbourhoods of $\mathfrak{q}$, directed by reverse inclusion. Suppose that every $V \in \mathcal{U}$ admits an ind-smooth algebra B_V over its coordinate ring and an integer d_V with

$$0 \neq x_V \in \pi_{d_V}(NK(B_V)_p^\wedge[1/p]).$$

For a separated finitely presented V -scheme T , put

$$\Phi_V(T) = \Sigma^{-d_V}(NK(T_{B_V})_p^\wedge[1/p]).$$

These are presheaves of rational spectra, and $x_V \in \pi_0 \Phi_V(V)$.

For each $U \in \mathcal{U}$, let $\mathcal{D}_U$ be the category of separated finitely presented U -schemes, and define

$$G_U(T) = \operatorname{colim}_{\substack{W \in \mathcal{U} \\ W \subseteq U}} \prod_{\substack{V \in \mathcal{U} \\ V \subseteq W}} \Phi_V(T_V). \quad (7.1)$$

The transition maps discard coordinates. For $U' \subseteq U$, the subfamily $W \subseteq U'$ is cofinal, giving a natural equivalence $G_U(T) \simeq G_{U'}(T_{U'})$, compatible with further restriction. Finite-presentation descent identifies

$$\operatorname{colim}_{U \in \mathcal{U}} \mathcal{D}_U \simeq \mathcal{D}_{\operatorname{Spec} A_{\mathfrak{q}}} :$$

objects and morphisms descend, and equality of morphisms holds at a sufficiently small stage [18, Tag 01ZM]. The compatible presheaves G_U therefore define a presheaf G over $A_{\mathfrak{q}}$, by the universal property of this colimit of categories.

Homotopy groups commute with products of spectra and filtered colimits. Hence

$$\pi_0 G(\operatorname{Spec} A_{\mathfrak{q}}) = \operatorname{colim}_{U \in \mathcal{U}} \prod_{V \subseteq U} \pi_0 \Phi_V(V).$$

The family (x_V) gives a nonzero element: every tail contains the coordinate $V = U$, where the chosen element is nonzero.

We verify the hypotheses of Proposition 5.1. If T is a regular scheme over A_q , it has a regular model after shrinking the base, by Lemma 6.1. Lemma 3.4 then makes every factor in a sufficiently small tail zero. Therefore $G(T) = 0$.

A relation in $\mathcal{R}_C$ descends to a neighbourhood by Lemma 6.1. Empty and disjoint-union relations and Nisnevich squares are annihilated by every Φ_V . For a proper relation, the lemma preserves integral cdh acyclicity, properness, and regularity of the generic fibres, so Proposition 4.2 gives the same conclusion. Products and filtered colimits of spectra commute with the finite total-fibre construction. Consequently G annihilates each relation in $\mathcal{R}_C$.

Finally, a finite flat surjection $Y \rightarrow T$ whose direct image is free of positive constant rank descends with a chosen basis to a neighbourhood. On each subsequent coordinate, the transfer identity in Lemma 3.4 makes pullback split injective on rational homotopy groups. Products and filtered colimits of abelian groups preserve injectivity, so $\pi_j G(T) \rightarrow \pi_j G(Y)$ is injective for every j .

The Noetherian local ring A_q is quasi-excellent and finite-dimensional by the height theorem, and its generic fibre is regular. Proposition 5.1 gives $G(\operatorname{Spec} A_q) = 0$, contradicting the nonzero class above. This proves the proposition. $\square$

Theorem 7.2. *Let A be Noetherian and quasi-excellent, with $A[1/p]$ regular. For every ind-smooth A -algebra B and every $m \geq 1$,*

$$N_m K(B)_p^\wedge[1/p] = 0.$$

Proof. First suppose that A is a $\mathbb{Z}_{(p)}$ -algebra. Quasi-compactness and Proposition 7.1 give a finite cover

$$\operatorname{Spec} A = D(f_1) \cup \cdots \cup D(f_r).$$

If B is ind-smooth over A , then B_{f_i} and its further localizations at the f_j are ind-smooth over A_{f_i} . The proposition gives vanishing on every intersection in the induced cover of $\operatorname{Spec} B$. Each operation defining $N_m K(-)_p^\wedge[1/p]$ preserves finite limits. Finite Mayer–Vietoris induction therefore gives vanishing on $\operatorname{Spec} B$.

For general A , replace it by $A \otimes_{\mathbb{Z}} \mathbb{Z}_{(p)}$. This is a Noetherian quasi-excellent localization with regular generic fibre, and the corresponding base change of B is ind-smooth. Lemma 3.2 identifies the ordinary Nil spectra before and after this replacement. The preceding case completes the proof. $\square$

8 A common exponent

Theorem 7.2 and Lemmas 3.2–3.3 show that each group $N_m K_j(B)$ has bounded p -exponent. We now make the bound independent of the algebra, the number of variables, and the degree below a fixed upper bound.

Lemma 8.1. *Fix a ring A and an integer N . Suppose that $NK_N(B)$ is annihilated by a power of p for each ind-smooth A -algebra B . Then one power of p annihilates all these groups.*

Proof. Ind-smooth A -algebras are closed under finite products and filtered colimits. For products, use finite products of smooth finitely presented stages, which correspond to finite disjoint unions of smooth affine schemes. For filtered colimits, one may use the finite-presentation criterion for ind-smoothness: every map from a finitely presented A -algebra to the algebra factors through a smooth finitely presented A -algebra. Such a map to a

filtered colimit factors through one stage, where the criterion applies. This criterion also shows that the category of smooth finitely presented A -algebras mapping to the algebra is filtered: finite diagrams and equalities are encoded by finitely presented A -algebras.

Suppose that no common exponent exists. Choose ind-smooth A -algebras B_i , $i \geq 1$, such that

$$p^i NK_N(B_i) \neq 0.$$

Write $f_i : A \rightarrow B_i$ for the structure maps and form the filtered system

$$\begin{aligned} C_r &= B_1 \times \cdots \times B_r \times A, \\ C_r &\longrightarrow C_{r+1}, \\ (b_1, \dots, b_r, a) &\longmapsto (b_1, \dots, b_r, f_{r+1}(a), a). \end{aligned}$$

Its colimit C is ind-smooth over A . For a fixed i , the i -th coordinate idempotent is preserved from stage i onwards. It gives a unital product decomposition

$$C \simeq B_i \times C'_i$$

for some ring C'_i . Since K -theory preserves finite products, $NK_N(B_i)$ is a direct summand of $NK_N(C)$. By hypothesis, $p^e NK_N(C) = 0$ for some e . This annihilates every $NK_N(B_i)$, contradicting the choice of any $i \geq e$. $\square$

Lemma 8.2. *For every ring B and every $m \geq 1$, the spectrum $\Sigma N_m K(B)$ is naturally a direct summand of $N_m K(B[u, u^{-1}])$.*

Proof. The natural nonconnective Bass–Heller–Swan decomposition is

$$K(R[u, u^{-1}]) \simeq K(R) \oplus \Sigma K(R) \oplus NK(R) \oplus NK(R);$$

see [14, Theorem 0.1 and Example 0.3]. Apply it to $R = B[t_1, \dots, t_m]$ and $R = B$, and take fibres along zero evaluation. Naturality gives the asserted summand. $\square$

Proof of Theorem 1.2. For each ind-smooth A -algebra B , Lemma 3.2 and Theorem 7.2 give

$$N_m K(B)[1/p] = 0, \quad N_m K(B)_p^\wedge[1/p] = 0.$$

Lemma 3.3 therefore gives bounded p -exponent in every homotopy group. Fix $N \in \mathbb{Z}$. By Lemma 8.1, there is $e = e(A, p, N)$ with

$$p^e NK_N(B) = 0 \quad \text{for every ind-smooth } B/A.$$

Polynomial extensions of ind-smooth algebras are ind-smooth. The direct-sum decomposition in Lemma 3.1 therefore gives $p^e N_m K_N(B) = 0$ for every m .

For $j \leq N$, iterate the retraction in Lemma 8.2 $N - j$ times. It exhibits $N_m K_j(B)$ as a direct summand of

$$N_m K_N(B[u_1^{\pm 1}, \dots, u_{N-j}^{\pm 1}]).$$

This Laurent extension is still ind-smooth over A , so its indicated group is annihilated by the same p^e . This proves the theorem. $\square$

Proof of Theorem 1.1. For a union $T = U \cup V$ of quasi-compact open subschemes, Zariski Mayer–Vietoris gives an exact sequence

$$N_m K_{j+1}(U \cap V) \longrightarrow N_m K_j(T) \longrightarrow N_m K_j(U) \oplus N_m K_j(V).$$

Suppose that p^a and p^b annihilate the groups for U and V , respectively, in all degrees at most N , uniformly in m , and that p^c annihilates the groups for $U \cap V$ in all degrees at most $N + 1$, uniformly in m . The exact sequence then gives the uniform bound

$$p^{\max(a,b)+c} N_m K_j(T) = 0 \quad (m \geq 1, j \leq N). \quad (8.1)$$

First let W be a quasi-compact open subscheme of an affine open of X . Choose a finite cover of W by distinguished opens of that affine scheme. All finite intersections in this cover are distinguished affine opens. Their coordinate rings are Noetherian and quasi-excellent and become regular after inverting p , so Theorem 1.2 gives the required uniform bounds on them. Induction on the number of distinguished opens, using (8.1), gives such a bound on W for every integer N . At each induction step the intersection is covered by fewer distinguished opens, and the degree bound increases only from N to $N + 1$.

Now choose a finite affine cover $X = U_1 \cup \cdots \cup U_r$ and put $V_i = U_1 \cup \cdots \cup U_i$. Since X is Noetherian, $V_{i-1} \cap U_i$ is a quasi-compact open subscheme of the affine scheme U_i . The preceding paragraph applies to every such intersection. Apply (8.1) successively to $V_i = V_{i-1} \cup U_i$. There are only finitely many steps, each using bounds independent of m and uniform in all degrees below its specified upper bound. They produce one exponent $e(X, p, N)$ with the asserted properties. $\square$

A Extension from schemes with regular generic fibre

Let S be a finite-dimensional Noetherian quasi-excellent $\mathbb{Z}_{(p)}$ -scheme such that $S[1/p]$ is regular, and put $U = S[1/p]$. Let $\mathcal{D}$ be the category of separated schemes of finite presentation over S , and let $\mathcal{C} \subset \mathcal{D}$ consist of those T for which T_U is regular. The proper relations used below are those of Definition 4.1: finite geometric relations which are annihilated by every spectrum-valued cdh sheaf, whose vertices lie in $\mathcal{C}$ and are proper over a common object of $\mathcal{D}$.

Theorem A.1 (Finite-dimensional extension). *Let $F : \mathcal{C}^{\text{op}} \rightarrow \text{Sp}_{\mathbb{Q}}$ be a presheaf of rational spectra. Suppose that F takes the empty scheme to zero, takes finite disjoint unions to products, satisfies Nisnevich descent on squares in $\mathcal{C}$, and annihilates every finite proper geometric relation in $\mathcal{C}$. Then F extends to a rational spectrum-valued cdh sheaf on $\mathcal{D}$. Moreover, every object of $\mathcal{D}$ admits a cdh cover by objects of $\mathcal{C}$.*

The geometric input is compatible cubical resolution in characteristic zero, in the form proved by Murayama [16, Theorem 4.3.4 and Proposition 4.4.7]. We extend the resulting proper diagrams from U to S and add the special fibre to their centres. This produces finite resolutions by objects of $\mathcal{C}$. To extend F , we show that these resolutions identify the stable categories obtained by imposing its descent relations on $\mathcal{C}$ and the cdh relations on $\mathcal{D}$. For the quotient comparison, a compatible resolution of a single map of complexes and a homotopy pushout give the factorization criterion for full faithfulness. Resolutions of the geometric generators then give essential surjectivity. This follows the cubical extension strategy of Guillén–Navarro Aznar [10]; the extension over S is proved below using resolutions of the characteristic-zero fibre.

A.1 Compatible finite resolutions

Write $\mathcal{D}_{\text{red}}$ and $\mathcal{C}_{\text{red}}$ for the full subcategories of reduced objects. Fibre products in $\mathcal{D}_{\text{red}}$ are reductions of ordinary fibre products. A category with finitely many morphisms is *orderable*

if its objects admit an ordering in which every nonidentity arrow points strictly forward. Parallel arrows and relations between paths are allowed. For a diagram $X : I \rightarrow \mathcal{D}_{\text{red}}$ indexed by such a category, put

$$d(X) = \max_{i \in I} \dim(X_{i,U}), \quad \dim(\emptyset) = -1.$$

Lemma A.2 (Proper models of generic diagrams). *Let $X : I \rightarrow \mathcal{D}_{\text{red}}$ be a finite orderable diagram. Suppose that $Y_U \rightarrow X_U$ is a morphism of diagrams of reduced schemes over U , proper at every vertex. There is a vertexwise proper morphism $Y \rightarrow X$ of diagrams in $\mathcal{D}_{\text{red}}$ whose restriction to U is the given morphism. Each Y_i may be chosen with schematically dense generic fibre $Y_{i,U}$.*

Proof. For each i , Nagata compactification gives a proper model of $Y_{i,U}$ over X_i ; see [18, Tag 0F3T]. The open immersion from $Y_{i,U}$ into the generic fibre of this compactification is proper, because its source is proper over $X_{i,U}$ and its target is separated over $X_{i,U}$. It is therefore an open and closed immersion. Taking the reduced closure of $Y_{i,U}$ gives a proper model $P_i \rightarrow X_i$ with $P_{i,U} = Y_{i,U}$ and schematically dense generic fibre. If $Y_{i,U}$ is empty, take $P_i = \emptyset$.

Construct the diagram in reverse order, so that the models Y_j are already defined for every nonidentity arrow $a : i \rightarrow j$. The scheme

$$P_i \times_{X_i} \prod_{\substack{a:i \rightarrow j \\ a \neq \text{id}}} (X_i \times_{X_j} Y_j)$$

is proper over X_i . Over U , the prescribed arrows out of $Y_{i,U}$, together with its identity map to $P_{i,U}$, define a closed graph in this product. Let Y_i be its reduced closure. Projection gives all required arrows $Y_i \rightarrow Y_j$.

Two paths which agree in I now give maps $Y_i \rightarrow Y_k$ agreeing on $Y_{i,U}$. Their equalizer is closed, since Y_k is separated over S , and contains a schematically dense subscheme. The maps therefore agree on Y_i , as required. $\square$

For a finite orderable diagram in characteristic zero, the very weak resolution theorem and its two-resolution construction give a regular diagram R_U , a closed subdiagram $D_U \subset X_U$, and $E_U = (D_U \times_{X_U} R_U)_{\text{red}}$, with abstract blowup squares at every vertex and

$$\max_i \dim D_{i,U} < d(X), \quad \max_i \dim E_{i,U} < d(X);$$

see [16, Theorem 4.3.4 and Proposition 4.4.7]. The strict decrease of the largest dimension in the diagram makes the following simultaneous construction finite.

Lemma A.3 (Compatible cubical resolutions). *Every finite orderable diagram $X : I \rightarrow \mathcal{D}_{\text{red}}$ admits compatible augmented finite cubical resolutions $R_{i,\bullet} \rightarrow X_i$, with a common finite cubical index, such that:*

1. *every nonaugmented vertex lies in $\mathcal{C}_{\text{red}}$ and is proper over its corresponding X_i ;*
2. *all arrows and path identities in I are preserved strictly;*
3. *every spectrum-valued cdh sheaf H satisfies*

$$H(X_i) \simeq \text{holim}_{\alpha} H(R_{i,\alpha}).$$

This equivalence also holds after any reduced base change.

The base-changed vertices in the last assertion may have singular generic fibres.

Proof. Induct on $d(X)$. If $d(X) = -1$ or 0 , all generic fibres are regular, so the trivial resolutions suffice. Otherwise apply the characteristic-zero construction above and extend $R_U \rightarrow X_U$ by Lemma A.2. Writing $V = S \setminus U$, set

$$D_i = (\overline{D_{i,U}} \cup X_{i,V})_{\text{red}}, \quad E_i = (D_i \times_{X_i} R_i)_{\text{red}}.$$

The original arrows preserve both the closures of the generic centres and the vertical fibres. Their restrictions to the reduced schemes D_i therefore factor through the corresponding D_j . The D_i , and consequently the E_i , form diagrams.

The complement $X_i \setminus D_i$ lies in U , where $R_i \rightarrow X_i$ is an isomorphism over this complement. Hence

$$\begin{array}{ccc} E_i & \longrightarrow & R_i \\ \downarrow & & \downarrow \\ D_i & \longrightarrow & X_i \end{array}$$

is a reduced abstract blowup square. Each R_i lies in $\mathcal{C}_{\text{red}}$, and the largest generic-fibre dimension in the residual diagram $E \rightarrow D$ is smaller than $d(X)$. Apply induction to this entire diagram, indexed by $\{0 \rightarrow 1\} \times I$, and combine its resolutions with the displayed squares by cubical reduction. This gives a common finite cubical shape while retaining every map and path identity in I . Properness is preserved by base change and composition.

A cdh sheaf takes each abstract blowup square to a cartesian square. Repeated finite limits give the asserted equivalence. The same argument applies after reduced base change, which preserves these squares. $\square$

The proper modification and closed centre in an abstract blowup square form a cdh cover. Composing these covers through the finite construction shows that every object of $\mathcal{D}_{\text{red}}$ is cdh covered by objects of $\mathcal{C}_{\text{red}}$.

Lemma A.4 (Finite common comparisons). *Two compatible resolutions of the same finite orderable diagram admit a compatible finite iterated comparison*

$$A_{\bullet} \longleftarrow Z_{\bullet} \longrightarrow B_{\bullet}$$

of geometric total complexes. At each original vertex, the two comparison cones are finite proper geometric relations in $\mathcal{C}_{\text{red}}$, with common base the original scheme.

Proof. Write the two resolutions over X_i as $A_{i,\alpha}$ and $B_{i,\beta}$. Apply Lemma A.3 to the finite diagram

$$(A_{i,\alpha} \times_{X_i} B_{i,\beta})_{\text{red}}.$$

Denote its resolution vertices by $Z_{i,\alpha,\beta,\gamma}$ and totalize over the three finite indices. The augmentations give the two comparison maps, compatibly in i . Every vertex belongs to $\mathcal{C}_{\text{red}}$ and is proper over X_i .

For a spectrum-valued cdh sheaf, the finite limit in γ recovers the values on the reduced fibre products. The limit in β then recovers the values on $A_{i,\alpha}$, by the base-change assertion of Lemma A.3. Taking the remaining limit proves that the first comparison is an equivalence. Interchanging α and β proves the same for the second comparison. $\square$

A.2 Stable quotients and finite complexes

For a small category $\mathcal{E}$, let $\mathcal{A}(\mathcal{E})$ be the stable subcategory of $\text{Fun}(\mathcal{E}^{\text{op}}, \text{Sp}_{\mathbb{Q}})$ generated by the rational representables $\mathbb{Q}[T]$ under finite sums, shifts, and cofibres. Its idempotent completion is the subcategory of compact presheaves. Here $\mathbb{Q}[T](V)$ is the Eilenberg–Mac Lane spectrum of the free rational vector space on $\text{Hom}_{\mathcal{E}}(V, T)$. These representables have mapping spectra concentrated in degree zero, with

$$\pi_0 \text{Map}(\mathbb{Q}[T], \mathbb{Q}[V]) = \mathbb{Q}[\text{Hom}_{\mathcal{E}}(T, V)].$$

Using the equivalence between rational spectra and complexes of rational vector spaces, and the resulting model for diagram categories [15, Theorem 7.1.2.13 and Proposition 1.3.4.25], the homotopy category of $\mathcal{A}(\mathcal{E})$ is $K^b(\mathbb{Q}\mathcal{E})$. Here $\mathbb{Q}\mathcal{E}$ is the free additive rational category on $\mathcal{E}$. Bounded complexes of these projective representables compute the finite constructions and their maps. We use cohomological grading: differentials have degree 1, and homotopies have degree -1 .

Apply this construction to $\mathcal{C}_{\text{red}}$ and $\mathcal{D}_{\text{red}}$, writing the resulting categories as $\mathcal{A}_{\mathcal{C}}$ and $\mathcal{A}_{\mathcal{D}}$. The geometric inclusion identifies $\mathcal{A}_{\mathcal{C}}$ with a full stable subcategory of $\mathcal{A}_{\mathcal{D}}$. In $\mathcal{A}_{\mathcal{D}}$, let $\mathcal{R}_{\mathcal{D}}$ be the thick stable subcategory generated by Nisnevich squares, reduced abstract blowup squares, the empty scheme, and finite disjoint union relations. In $\mathcal{A}_{\mathcal{C}}$, define $\mathcal{R}_{\mathcal{C}}$ using the empty and disjoint union relations, Nisnevich squares in $\mathcal{C}_{\text{red}}$, and the rational linearizations of its finite proper geometric relations. Here “thick” means closed under finite sums, shifts, cofibres, and retracts which exist in the ambient category.

Take their Verdier quotients in small stable categories. Their universal property makes exact functors out of a quotient equivalent to exact functors annihilating the specified relations. Their homotopy categories are the corresponding triangulated Verdier quotients [4, Proposition 5.14]. We use the same notation $\mathcal{R}_{\mathcal{C}}, \mathcal{R}_{\mathcal{D}}$ for the induced thick subcategories in the homotopy categories; thus, for example,

$$\text{Ho}(\mathcal{A}_{\mathcal{D}}/\mathcal{R}_{\mathcal{D}}) \simeq K^b(\mathbb{Q}\mathcal{D}_{\text{red}})/\mathcal{R}_{\mathcal{D}}.$$

The compact-generation argument in the proof of Lemma 6.1, now with rational coefficients, shows that a finite complex killed by sheafification belongs to $\mathcal{R}_{\mathcal{D}}$. Every generator of $\mathcal{R}_{\mathcal{C}}$ is such a complex. The geometric inclusion therefore induces an exact functor

$$J : \mathcal{A}_{\mathcal{C}}/\mathcal{R}_{\mathcal{C}} \longrightarrow \mathcal{A}_{\mathcal{D}}/\mathcal{R}_{\mathcal{D}}. \quad (\text{A.1})$$

Lemma A.5. *Replace every vertex of a generating relation of $\mathcal{R}_{\mathcal{D}}$ by compatible resolutions from Lemma A.3. The resulting total complex belongs to $\mathcal{R}_{\mathcal{C}}$.*

Proof. For an abstract blowup square, all replacement vertices are proper over its original lower-right vertex. The square and the vertexwise augmentations are cdh equivalences. Their total replacement is therefore a finite proper geometric relation in $\mathcal{C}_{\text{red}}$.

For a Nisnevich square, choose a resolution of its lower-right vertex and pull it back along the open and étale maps in the square. Étale base change preserves reducedness and regularity of the generic fibres. Every cubical layer is now a Nisnevich square in $\mathcal{C}_{\text{red}}$. The filtration by cubical degree shows that its total relation belongs to $\mathcal{R}_{\mathcal{C}}$.

Compare any other compatible replacement with this one using Lemma A.4, simultaneously at all four vertices. The cone of the comparison of total relations has a finite filtration whose graded pieces are shifts of the vertexwise comparison cones. Each of these lies in $\mathcal{R}_{\mathcal{C}}$, so the other replacement does too. For a disjoint union, use the disjoint union of the chosen component resolutions and the same comparison argument. The empty relation is immediate. $\square$

A.3 Equivalence of the descent quotients

Proposition A.6. *The geometric inclusion induces an equivalence of idempotent-complete stable categories*

$$(\mathcal{A}_{\mathcal{C}}/\mathcal{R}_{\mathcal{C}})^{\text{idem}} \simeq (\mathcal{A}_{\mathcal{D}}/\mathcal{R}_{\mathcal{D}})^{\text{idem}}.$$

Proof. We work first in the homotopy categories. The factorization criterion for Verdier quotients [8, Lemma A.5; arXiv Lemma 12.5] says that J is fully faithful if every morphism

$$f : P \longrightarrow R, \quad P \in \mathcal{A}_{\mathcal{C}}, \quad R \in \mathcal{R}_{\mathcal{D}},$$

factors through an object of $\mathcal{R}_{\mathcal{C}}$. We verify this by resolving a single map of bounded complexes.

Since $\mathcal{R}_{\mathcal{D}}$ is thickly generated by the elementary relations, choose maps

$$R \xrightarrow{i} L \xrightarrow{r} R, \quad ri = \text{id}_R,$$

where L is a finite iterated cone of shifts and finite sums of those relations; see [18, Tag 0FX4]. Choose the cone model of L with its finite filtration by shifted elementary relations, and represent $if : P \rightarrow L$ by a chain map.

Give a separate label to each occurrence of a scheme in P and L . Put the occurrences in P before those in L , and order each complex by increasing cohomological degree. The geometric entries of the differentials and of if then form a finite acyclic graph. Identify paths with the same endpoints whenever they give the same scheme morphism. The resulting category is finite and orderable. Lemma A.3 resolves this diagram compatibly. Because morphism spaces are free rational vector spaces on scheme morphisms, preserving these path equalities preserves the linear identities $d^2 = 0$ and the chain map identity. Totalizing with differential $d_{\text{out}} + (-1)^n d_{\text{in}}$, where n is the original degree, gives an augmentation square

$$\begin{array}{ccc} \tilde{P} & \longrightarrow & \tilde{L} \\ a \downarrow & & \downarrow \\ P & \xrightarrow{if} & L. \end{array} \tag{A.2}$$

Replacement preserves the filtration of L . Its graded pieces are compatible replacements of elementary relations, so Lemma A.5 gives $\tilde{L} \in \mathcal{R}_{\mathcal{C}}$. Every scheme occurring in P already lies in $\mathcal{C}_{\text{red}}$. Its augmentation cone is therefore a finite proper geometric relation in $\mathcal{C}_{\text{red}}$, over that scheme. Filtering by the original degrees gives $\text{Cone}(a) \in \mathcal{R}_{\mathcal{C}}$.

Form the homotopy pushout

$$Q = P \amalg_{\tilde{P}} \tilde{L} \quad \text{in } \mathcal{A}_{\mathcal{C}}.$$

The cofiber sequence

$$\tilde{L} \longrightarrow Q \longrightarrow \text{Cone}(a)$$

shows that $Q \in \mathcal{R}_{\mathcal{C}}$. The square (A.2) induces $Q \rightarrow L$, and $ri = \text{id}_R$ identifies f with the composite

$$P \longrightarrow Q \longrightarrow L \xrightarrow{r} R.$$

This proves the required factorization, hence full faithfulness.

For every $T \in \mathcal{D}_{\text{red}}$, Lemma A.3 gives a finite complex over $\mathcal{C}_{\text{red}}$ whose augmentation to $\mathbb{Q}[T]$ is invertible in the right quotient. The essential image of J contains these generators and, by full faithfulness and exactness, is closed under finite sums, shifts, and cones. Thus J is essentially surjective. The resulting equivalence on homotopy categories is an equivalence of stable categories [4, Proposition 5.10 and Corollary 5.11]. Taking idempotent completions proves the proposition. $\square$

A.4 Extension of the presheaf

Proof of Theorem A.1. Regard the restriction of F as an object of $\mathrm{Fun}(\mathcal{C}_{\mathrm{red}}^{\mathrm{op}}, \mathrm{Sp}_{\mathbb{Q}})$. Mapping into F gives an exact contravariant functor on $\mathcal{A}_{\mathcal{C}}$, whose value on $\mathbb{Q}[T]$ is $F(T)$ by Yoneda.

The functor determined by the restriction of F annihilates $\mathcal{R}_{\mathcal{C}}$, so it factors through the stable quotient. The proof of Proposition A.6 identifies the quotients already before idempotent completion. Transport the functor across this equivalence and restrict to the geometric representables in $\mathcal{A}_{\mathcal{D}}$. This yields a presheaf $\overline{F}$ on $\mathcal{D}_{\mathrm{red}}$ which annihilates the finite cdh square relations. It is a cdh sheaf by [9, Propositions 2.1.5–2.1.6]. Its restriction to $\mathcal{C}_{\mathrm{red}}$ is naturally equivalent to that of F , since the quotient equivalence is induced by the geometric inclusion.

For $T \in \mathcal{C}$, the proper cdh equivalence $T_{\mathrm{red}} \rightarrow T$ is an allowed proper relation, so $F(T) \simeq F(T_{\mathrm{red}})$. Nilpotent thickenings are cdh covers, and reduction takes the generating cdh squares to their reduced versions. Therefore

$$T \longmapsto \overline{F}(T_{\mathrm{red}})$$

defines a cdh sheaf on the full category $\mathcal{D}$. The preceding equivalences identify its restriction to $\mathcal{C}$ with F . Finally, compose $T_{\mathrm{red}} \rightarrow T$ with the cover supplied by Lemma A.3. Its members belong to $\mathcal{C}$, which proves the covering assertion. $\square$

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